

JOSHUA NWACHINEMERE

AI Engineer | Applied AI, ML Systems, Automation

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFILE

AI Engineer and VolyxAI founder building practical AI, automation, and backend systems with Python. Work includes machine-learning pipelines, local retrieval, event-driven workflows, real-time services, and API integrations. Focused on systems that validate model output, preserve execution context, and keep people responsible for consequential actions.

PROFESSIONAL EXPERIENCE

Founder & AI Automation Engineer | Nov 2025 - Present

VolyxAI

- Designing early-stage operations workflows that connect conversation intake to validation, approval, and context-rich handoff steps.
- Built n8n workflow prototypes that connect language models with calendars, email, webhooks, and structured business rules in controlled test environments.
- Prototyped a voice-led scheduling flow using Vapi, ElevenLabs, n8n, Google Calendar, and email confirmation integrations.
- Defined input validation, retry limits, idempotency controls, and human approval requirements for consequential workflow actions.

Key tools: Python, n8n, TypeScript, REST APIs, webhooks, Vapi, ElevenLabs, Azure AI Foundry

Software & Automation Developer | Jan 2021 - Present

Freelance / Self-Employed

- Built Python automation and backend utilities for data processing, Telegram bots, API integrations, and operational tooling.
- Developed stream-oriented utilities for processing and splitting large text datasets without loading complete files into memory.
- Created asynchronous Telegram integrations for wallet analysis, malware scanning, user tracking, pagination, caching, and admin reporting.
- Built public Python tools for DNS intelligence, multi-source wallet analysis, and malware-report delivery.

Key tools: Python, Async I/O, Telegram Bot API, aiohttp, SQL, Docker, REST APIs

Threat Detection & Response Lab | Jul 2025 - Aug 2025

Capstone IT Staffing | Remote

- Deployed Suricata IDS on Ubuntu, wrote custom Snort rules, and automated alert enrichment with Python and public OSINT sources.
- Built a Python DNS intelligence CLI with asynchronous resolution, WHOIS lookups, trace output, and JSON export for investigations.

Key tools: Python, Suricata, Snort, OSINT, DNS, Linux

SELECTED PROJECTS

Noughtline | Real-Time Multiplayer System

React, Express, Socket.IO, SQLite, Paystack

- Built signed guest sessions, server-authoritative moves, reconnect and forfeit handling, transactional match settlement, and an append-only currency ledger.
- Added one-time crediting for verified Paystack references and automated API, economy, room-state, and two-client Socket.IO tests.
- Live: <https://noughtline.onrender.com> | Source: <https://github.com/dk3yyyy/Noughtline>

Football Predictor | Experimental ML Pipeline

Python, XGBoost, scikit-learn, FastAPI, Streamlit, SQLAlchemy

- Built data scrapers, temporal feature engineering, XGBoost outcome and Poisson goal models, a FastAPI service, and a Streamlit dashboard.
- Uses chronological train, calibration, and test partitions with probability calibration; no performance claim is published while evaluation work continues.
- Source: https://github.com/dk3yyyy/football_predictor

Local Review RAG | Local Retrieval Experiment

Python, Ollama, LangChain, Chroma, Llama 3.2

- Embedded restaurant reviews locally, stored vectors in Chroma, retrieved the five closest records, and supplied that context to a local language model.
- Source: https://github.com/dk3yyyy/local_AI_agent

VirusTotal Telegram Bot | Security API Integration

Python, Pyrogram, aiohttp, VirusTotal API

- Built asynchronous file, URL, and hash scanning with local SHA-256 lookup, VirusTotal large-file upload handling, bounded retry backoff, and short-lived report caching.
- Source: <https://github.com/dk3yyyy/VirusTotal-Telegram-Bot>

CORE SKILLS

- Languages: Python, JavaScript, TypeScript, SQL
- AI / ML: XGBoost, scikit-learn, pandas, Ollama, LangChain, Chroma, retrieval workflows, structured outputs
- Backend: FastAPI, Express, Socket.IO, SQLAlchemy, SQLite, PostgreSQL, REST APIs, webhooks
- Automation: n8n, approval gates, retries, idempotency, Telegram integrations, Google Calendar and email workflows
- Infrastructure: Docker, GitHub Actions, Linux, Microsoft Azure, Azure AI Foundry, GitHub Pages, Cloudflare Pages
- Security: Suricata, Snort, VirusTotal API, OSINT, DNS analysis, credential handling

CERTIFICATIONS

- Google AI Professional Certificate | Coursera
- Google Cybersecurity Professional Certificate | Coursera

EDUCATION

Bachelor of Technology (BTech), Mathematics | 2016 - 2021

Federal University of Technology, Owerri (FUTO)